AI-Based Multi-Domain Detection System

ECG • Brain-Tumor MRI • EEG • Satellite Imagery

- A Flask web app that runs deep-learning analysers behind a simple web UI and generates a PDF report for each analysis.
- This document explains the project from absolute basics to end — what it is, why it exists, how each module works, how it is trained, what the numbers mean, its limitations, and how to demo it.
- Important: this is a RESEARCH PROTOTYPE, not a medical device. Clinical decisions always need a qualified doctor.

1. What Is This Project? (Basics)

One app, four AI analysers

- You upload a file — an ECG heart signal, an MRI brain scan, an EEG brain recording, or a satellite image.
- The app analyses it with a trained neural network (TensorFlow/Keras).
- It shows the result on screen with plots/heatmaps AND gives a downloadable PDF report.
- It also stores studies in patient records (SQLite) with history and trends.
- Web pages: home (/) plus /ecg, /brain-tumor, /eeg, /satellite. Run with: .venv/bin/python app.py -> http://127.0.0.1:5050.

2. Problem Statement — Why This Exists

Manual analysis is slow, expert-dependent, and error-prone

- ECG: a cardiologist must inspect every heartbeat; rare ectopic beats are easy to miss in long recordings.
- MRI: radiologist shortages mean brain tumours need fast, consistent triage screening.
- EEG: seizures and early Alzheimer's signs hide inside hours of wavy brain signals no human can watch fully.
- Satellite: mapping forests, cities, water and farmland by hand does not scale to thousands of scenes.
- Goal of this project: fast AI screening that is HONEST — it refuses to guess when it has no valid model or no measurable signal.

3. Solution Overview — The 4 Modules

Same pattern everywhere: Upload -> AI -> Visuals -> PDF report

- ECG analyser (/ecg): accepts MIT-BIH CSV/TXT signal files or a PHOTO of an ECG strip; outputs NORMAL/ABNORMAL, beat distribution, heart rate, PR/QRS/QT intervals, rhythm, ST segment, axis, plus doctor notes.
- Brain-tumor MRI (/brain-tumor): accepts JPEG/PNG/DICOM MRI slice; outputs Glioma / Meningioma / No Tumor / Pituitary with confidence, Grad-CAM heatmap, tumour highlight and location.
- EEG analyser (/eeg): accepts EDF/BDF/SET/FIF or CSV brain recordings; outputs seizure and Alzheimer's screening with timeline plots and clinical notes.
- Satellite classifier (/satellite): accepts JPEG/PNG/TIFF scenes; outputs Forest / Urban / Water / Agriculture with NDVI vegetation index, land-breakdown shares and environmental health score.

4. Tech Stack (Beginner Glossary)

Plain pinned Python — no exotic infrastructure

- Frontend: HTML templates in templates/ (index, ecg, brain_tumor, eeg, satellite), Matplotlib waveform/timeline plots, Grad-CAM overlays for MRI.
- Backend: Flask in app.py — routes, file upload, model loading, report building. Uploads get safe uuid names and auto-expire after UPLOAD_TTL_SECONDS (default 24 h).
- AI: TensorFlow-Keras (1D-CNN for ECG, VGG16 transfer for MRI, EuroSAT CNN, scalogram CNNs for EEG), scikit-learn, SciPy, NumPy, pandas.
- Signal/image libs: OpenCV (ECG photo digitization), pydicom (DICOM MRI), Pillow, wfdb (MIT-BIH download).
- Reports & records: ReportLab PDF (reporting/pdf.py, study_reports.py); SQLite + files in records/; metadata in webapp/.
- Quality: pytest (137+ tests, model tests skip without checkpoints), Ruff lint, GitHub Actions CI.

5. System Architecture — How a Request Flows

Step-by-step journey of one upload

- Step 1 — User opens a page (e.g. /ecg), fills optional patient data, uploads a file or picks a sample.
- Step 2 — app.py validates the file, saves it under uploads/ with a generated name (ecg_upload.csv ...). Only generated names are servable — path traversal and stray .py/.keras files are blocked.
- Step 3 — Domain code processes input: ecg/beats.py segments R-peak-centred beats; vision/imageio.py loads MRI (16-bit windowed); eeg/ windows signals into scalograms.
- Step 4 — mlkit/registry.py validates the model's JSON sidecar card (task, class order, input shape, preprocessing). ANY mismatch = refused load, never silent wrong output.
- Step 5 — Model predicts; clinical modules (ecg/clinical.py, eeg/interpretation.py, vision/interpretation.py) add formula/range/verdict readings + precautions.
- Step 6 — Result page renders with plots; PDF report is built; if a patient is bound, the study is saved with verdict, model name/version and measurements for history/trends.

6. Module Deep-Dive: ECG (Heart)

1D-CNN + RR-interval context — AAMI 5-class beat classifier

- Input: numeric CSV/TXT at 360 Hz, or a photo of an ECG strip. Photos need a timebase: either you type the recording duration + paper speed (25/50 mm/s), or the grid is auto-detected; with neither, everything reports 'unavailable' and beats are NOT classified.
- Beat extraction: 280-sample window centred on R-peak (100 before + 180 after, ~0.78 s), per-beat z-score normalisation. Same function (ecg/beats.py) used in training AND the app so they cannot drift apart.
- Two inputs: (a) waveform (280,1) + (b) 4 RR ratios vs the record's own median — this is what lets the model catch supraventricular (S) beats, which are defined by being EARLY (0.66x median) not by shape.
- Classes: N normal, S supraventricular ectopic, V ventricular ectopic, F fusion, Q unclassifiable.
- Outputs: NORMAL/ABNORMAL + abnormal type, confidence, beat distribution/counts, HR, RR, QRS, PR, QT/QTc, SDNN, RMSSD, rhythm, ST, axis, signal quality (Good/Partial/Insufficient), interpretation + recommendation.
- Architecture lesson: the old GlobalAveragePooling1D head was translation-invariant (shifting beats 140 samples barely changed F1 0.285->0.279) — replaced with Flatten so position matters. Class weights capped at 20 so 6 Q-beats can't hijack training.

7. Module Deep-Dive: Brain-Tumor MRI

VGG16 transfer learning, 4 classes + Grad-CAM explainability

- Input: MRI slice as JPEG/PNG/DICOM. 16-bit and palette images are windowed by percentile (plain convert('RGB') would clip them white). Blank/flat images are refused with a clear message.
- Models: mri_vgg16.keras (224x224, ImageNet BGR mean subtraction) and mri_cnn.keras (128x128 scratch, /255). The app ranks candidates by each card's macro-F1 and serves the best validated one.
- Classes: Glioma, Meningioma, No Tumor, Pituitary — each with findings text + recommendation.
- Explainability: Grad-CAM heatmap (where the classifier looked) + tumour-region highlight + location (left/right/centre). Flat attention maps are reported as failures, not hidden.
- Known limitation (say this in Q&A;): tumour 'Estimated Area/Severity/Spread' from a 90th-percentile heatmap threshold is ~10% for every scan BY CONSTRUCTION — it is labelled Estimated because Grad-CAM is not segmentation. A BraTS-trained U-Net would make it a real measurement.

8. Module Deep-Dive: EEG (Brain Signals)

Scalogram CNNs for seizure + Alzheimer's screening

- Input: EDF/BDF/SET/FIF recordings or CSV exports (eeg/loaders.py, digitize.py).
- Method: signal split into 2-second windows with 50% overlap -> each window converted to a 64x64 time-frequency scalogram image (log min-max normalised) -> CNN classifies positive/negative.
- Two models: eeg_seizure (CHB-MIT, per-event labels; sensitivity 0.74, ROC-AUC 0.90 — catches seizures but precision 0.20, so a sensitive screen) and eeg_alzheimer (subject-level weak labels applied to every window — disclosed on the card; F1 0.60).
- Anti-leakage: splits are by SUBJECT/recording, never by window — the same person's windows can't appear in train and test.
- Outputs: timeline plot of seizure/Alzheimer probability over time, window counts, verdict, clinical notes, PDF report.

9. Module Deep-Dive: Satellite (Land Cover)

EuroSAT CNN — 10 classes collapsed to 4 display groups

- Input: satellite scene JPEG/PNG/TIFF (RGB only; the 13-band multispectral variant is NOT used).
- Model: EuroSAT 10-class CNN on 64x64 inputs (AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, SeaLake).
- Display grouping (by NAME, never index): Forest+HerbaceousVegetation -> Forest/Vegetation; Highway+Industrial+Residential -> Urban; River+SeaLake -> Water; AnnualCrop+Pasture+PermanentCrop -> Agriculture/Barren. Winner = highest summed group probability.
- Extras computed from the image: pseudo-NDVI = (Green-Red)/(Green+Red), land-breakdown shares (renormalised to 100%), environmental health score per winning group.
- Honesty: with no validated satellite model installed, the page uses a spectral RGB/NDVI heuristic AND says so in the method field — never presented as a CNN.

10. Honesty Guarantees (The Project's Signature Feature)

No plausible-looking number without proof — enforced by tests

- No fallback classifiers: predict_ecg() and predict_brain_tumor() RAISE when no validated model exists. Removed fallbacks: quantum-circuit classifier on unpickled cross-version scaler, sigmoid(std(signal)) as 'confidence', hardcoded '78.4% Glioma' by brightness.
- No unmeasured class names: any beat class with card F1 < MIN_REPORTABLE_F1 (0.30) is never named in UI/PDF. Currently S (0.26), F (0.008), Q (0.0) are suppressed into 'Other / not reliably classified' but still count toward abnormal.
- Reports state absence: missing classification prints 'Not performed + reason', not dashes that look like missing data.
- Measurements are measured: ECG intervals come from the signal with an explicit timebase (px/mm from stated duration or detected grid); photo with no grid -> 'unavailable'. Never constants.
- Registry: every checkpoint needs a matching JSON card; VGG16-fed-/255 and tumour-model-serving-land-cover defects are now refused loads. All enforced by tests/test_no_fabricated_output.py.

11. Data, Training & Model Cards

Reproducible pipeline — SEED = 42, deterministic splits, SHA-256 manifests

- MIT-BIH Arrhythmia (ODC-By 1.0, via wfdb): AAMI classes N/S/V/F/Q; de Chazal DS1/DS2 INTER-PATIENT split (no record in two splits); paced records 102/104/107/217 excluded; ~50k beats per split; MLII lead, R-centred 280-sample windows.
- Brain MRI (Kaggle Nickparvar, CC0): Testing/ kept whole as test; 15% stratified val from Training; 187 byte-identical dupes removed pre-split; 4616/813/1584 images.
- EuroSAT RGB (MIT licence, via HuggingFace parquet): publisher's shards used as-is; 27,000 images 64x64; 16,200/5,400/5,400.
- EEG: CHB-MIT seizure events + Alzheimer's subjects; subject/recording-level splits; manifests in data/manifests/.
- Pipeline commands: scripts/download_raw.py -> prepare_mri/eurosat/ecg/eeg.py -> scripts/train_all.sh (EPOCHS=1 = smoke test). Each trainer writes model/.keras + model/.json card (task, classes, shape, preprocessing, metrics, date). ~350 MB checkpoints are NOT in git.

12. Results — Measured Performance

Only numbers written on model cards may be claimed

- ECG (strict inter-patient test): accuracy 0.858, macro-F1 0.400 overall, 0.667 over supported N/S/V. Per-class F1: N 0.920, V 0.819, S 0.261, F 0.002, Q 0.000. S recall 0.10->0.46 after RR context, but precision 0.18 — a sensitive screen, not a confident call.
- MRI: VGG16 transfer accuracy 0.927 / macro-F1 0.925 vs scratch CNN 0.734/0.711 — transfer learning wins clearly; the app serves the better one automatically.
- Satellite: accuracy 0.957 / macro-F1 0.957.
- EEG: seizure ROC-AUC 0.90 / F1 0.32 (finds events, many false alarms); Alzheimer F1 0.60 / ROC-AUC 0.68.
- Exam-ready line: published ~99% MIT-BIH figures use leaky beat-wise splits (same patient both sides) and are NOT comparable — the lower numbers here are the CORRECT inter-patient numbers.

13. Limitations — Say These Proudly in Q&A;

A research prototype, not a medical device

- ECG: F (42 beats) and Q (6 beats) have almost no training data — no feature engineering fixes that; capping weights just stops them corrupting N/V. S stays below the 0.30 naming bar.
- MRI: source has no patient IDs, so patient-level split is impossible — slices from one patient may span train/test, so treat 0.927 as an UPPER bound. Near-duplicates undetected.
- EEG: Alzheimer's labels are weak (subject label on every window); deployed QRS/seizure detectors add error not in metrics; single test set will overfit with repeated evaluation.
- Satellite: RGB only (no near-infrared); geographically adjacent tiles can split across train/test (spatial autocorrelation); classes mildly imbalanced.
- Ops: without checkpoints installed, pages degrade with clear errors instead of guessing; uploads auto-expire (privacy + disk).

14. How to Run & Demo It (End-to-End)

From zero to PDF report in five minutes

- Setup: python -m venv .venv; .venv/bin/pip install -r requirements.txt -r requirements-eeg.txt; PORT env overrides port 5050.
- Models: .venv/bin/pip install -r requirements-train.txt; .venv/bin/python scripts/download_raw.py; python scripts/prepare_ecg.py (+ prepare_mri, prepare_eurosat, prepare_eeg); scripts/train_all.sh — or drop existing .keras + .json cards into model/.
- Run: .venv/bin/python app.py -> open http://127.0.0.1:5050.
- Demo script: (1) /ecg -> upload static/samples/sample_*_ecg.csv -> point at waveform, NORMAL/ABNORMAL, interval table -> Download PDF. (2) /brain-tumor -> sample MRI -> prediction + confidence + Grad-CAM heatmap -> PDF. (3) /satellite -> scene -> group + NDVI + breakdown. (4) Show a missing-model error to prove honesty.
- Verify: .venv/bin/python -m pytest -q; python3 -m ruff check . (both in CI).

15. Project Structure & Future Work

Where everything lives, and what comes next

- Layout: app.py (Flask routes + model loading) | ecg/ (beats, digitize, delineate, parameters, clinical, quality) | eeg/ | vision/ (imageio, gradcam, interpretation) | mlkit/ (registry, manifest) | scripts/ (download/prepare/train) | reporting/ (pdf, study_reports, history, trends) | webapp/ (patients, metadata, study_binding) | tests/ (30+ files) | model/ (cards+checkpoints) | templates/ + static/ + uploads/ + reports/ + records/.
- Future work: BraTS U-Net for REAL tumour segmentation (replacing Estimated morphometry); improving S-beat precision; event-level EEG scoring; multispectral (13-band) satellite model; patient-level MRI splits; prospective clinical validation; deployment hardening (auth, audit log).
- Closing line for your presentation: 'Lower ECG numbers are the honest numbers — this system would rather say I don't know than invent a diagnosis.' Thank you — Questions?